

FACULTY OF ENGINEERING
B.E. II - Semester (AICTE) (Main) Examination, October 2021

Subject: Mathematics - II

Time: 2 Hours

Max. Marks: 70

- Note:** i) First Question is compulsory and answer any three questions from the remaining six questions.
 ii) Answers to each question must be written at one place only and in the same order as they occur in the question paper.
 iii) Missing data, if any, may suitably be assumed.

Answer any four questions from the following.

(4x4=16 Marks)

- 1 a Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 1 & 3 & -1 \\ 8 & 13 & 14 \end{bmatrix}$.
- b Solve $y(2xy + e^x)dx = e^x dy$.
- c Solve $(D^2 + 9)y = \sin 3x$.
- d Evaluate $\int_0^{\infty} e^{-5x} (1 - e^{-x})^7 dx$ in terms of beta function.
- e Find $L\{t^3 e^t + \sin^2 t\}$.
- f Find $L^{-1}\left\{\frac{1}{(s^2 + 1)(s^2 + 3)}\right\}$.
- g Evaluate $6P_3(x) + 4P_2(x) - 16P_1(x)$ as a polynomial of x .

(3x18=54 Marks)

- 2 (a) Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$.
- (b) Reduce the quadratic form $8x_1^2 + 7x_2^2 + 3x_3^2 + 12x_1x_2 + 4x_1x_3 - 8x_2x_3$ into canonical form.
- 3 (a) Solve $y(x+y)dx - x^2 dy = 0$.
- (b) Solve $y(2xy+1)dx + x(1+2xy-x^3y^3)dy = 0$.
- 4 (a) Solve $y'' + 4y = x \cos x$.
- (b) Solve $y'' + 2y' + y = e^{-x} \log x$ by the method of variation of parameters.
- 5 (a) Find the power series solution of the differential equation $y'' + 2xy' + y = 0$ about the origin.

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(b) Evaluate $\frac{d}{dx} [\operatorname{erf}(\alpha x)]$.

6 (a) Find $L \left\{ \int_0^t u e^{-u} \sin 4u \, du \right\}$.

(b) Find $L^{-1} \left\{ \frac{1}{s^3(s+2)} \right\}$.

7 (a) Find the orthogonal trajectories of the family of curves $y^3 + 3x^2y = c$ where c is arbitrary constant.

(b) Solve $x^2 y'' - xy' - 3y = x^2 \log x$.

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FACULTY OF ENGINEERING

**B.E. II - Semester (AICTE) (Main & Backlog) New) Examination,
September/ October - 2022**

Subject : MATHEMATICS-II

Time : 3 Hours

Max. Marks: 70

- Note:** (i) First question is compulsory and answer any four questions from the remaining six questions. Each Questions carries 14 Marks.
(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.
(iii) Missing data, if any, may be suitably assumed.

1. (a) If λ is an eigenvalue of a non-singular matrix A , show that $\frac{|A|}{\lambda}$ is an eigenvalue of $\text{Adj } A$.
 (b) Obtain the general solution of the differential equation $y = xy' + e^{-y'}$.
 (c) Find the second order differential equation for which e^x, e^{-x} are solutions.
 (d) Prove that $\text{erf}(x) + \text{erfc}(x) = 1$.
 (e) Find $L\{(\cos t - \sin t)^2\}$.
 (f) Find the matrix of the quadratic form $Q = 2(x^2 + xy + y^2)$.
 (g) Find a particular integral of $y'' + 2y' + y = \sin x$.
2. (a) Show that the system of equations $x - 3y - 8z + 10 = 0$, $3x + y - 4z = 0$, $2x + 5y + 6z - 13 = 0$ is consistent and solve the same.
 (b) Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 2 & 1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$.
3. (a) Find the general solution of $(x^3 + y^3) dx - xy^2 dy = 0$.
 (b) Solve the differential equation $xy(1 + xy^2) \frac{dy}{dx} = 1$.

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4. (a) Solve $\frac{d^3 y}{dx^3} - y = (e^x + e^{-x})^2$.

(b) Solve $x^2 y'' - 2xy' + 2y = \frac{1}{x}$.

5. (a) Prove that $\beta(m, n) = \beta(n, m)$ and $\beta(m+1, n) + \beta(n+1, m) = \beta(m, n)$.

(b) Find the power series solution of the differential equation $(1-x^2)y'' - 2xy' + 2y = 0$ about the origin.

6. (a) Evaluate $\int_0^{\infty} t^3 e^{-t} \sin t \, dt$ using Laplace transform.

(b) Apply convolution theorem to find $L^{-1} \left\{ \frac{1}{s(s^2-1)} \right\}$.

7. (a) Define rank of a matrix. Find all values of k such that the rank of the matrix

$A = \begin{pmatrix} k & -1 & 0 & 0 \\ 0 & k & -1 & 0 \\ 0 & 0 & k & -1 \\ -6 & 11 & -6 & 1 \end{pmatrix}$ is equal to 3.

(b) Find the orthogonal trajectories of the family of curves $\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1$, where λ is a parameter.

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FACULTY OF ENGINEERING**B.E. II - Semester (AICTE) (BACKLOG) Examination, March / April 2022****Subject: Mathematics - II
(Common for All Branches)****Time: 3 Hours****Max. Marks: 70**

- Note:** (i) First question is compulsory and answer any four questions from the remaining six questions. Each question carries 14 marks.
(ii) Answer to each question must be written in one place only and in the same order as they occur in the question paper.
(iii) Missing data, if any, may be suitably assumed.

- 1 (a) Find the rank of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 0 & 2 & 1 \end{pmatrix}$.
- (b) Find the values of a and b so that the differential equation $(x^2 + axy - 2y^2)dx + (y^2 - 4xy + bx^2)dy = 0$ is exact.
- (c) Find a differential equation of the form $ay'' + by' + cy = 0$ for which the functions $1, e^{-x}$ are solutions.
- (d) Define error and complementary functions.
- (e) Find $L^{-1}\left\{\frac{1}{s^2 + 2s + 2}\right\}$.
- (f) State Cayley-Hamilton theorem.
- (g) Solve $2x^2y'' + xy' - 6y = 0$.
- 2 (a) Determine whether the vectors $(1, 1, 0, 1)$, $(1, 1, 1, 1)$, $(4, 4, 1, 1)$, $(1, 0, 0, 1)$ are linearly dependent.
- (b) Reduce the quadratic form $Q = 3x^2 - 2y^2 - z^2 - 4xy + 12yz + 8xz$ into canonical form and find the nature of the quadratic form.
- 3 (a) Solve $(x^2y - 2xy^2)dx - (x^3 - 3x^2y)dy = 0$.
- (b) Find the general solution of the Riccati's equation $y' = 4xy^2 + (1 - 8x)y + 4x - 1$, if $y = 1$ is a particular solution.
- 4 (a) Solve $y'' + 2y' + 2y = x^3 + 6\cos^2 x$.
- (b) Apply the method of variation of parameters to solve $y'' + y = \tan x$.
- 5 (a) Evaluate the following integrals using Beta and Gamma functions.
- (i) $\int_0^{\infty} 2^{-4x^2} dx$ (ii) $\int_{-1}^1 (1-x^2)^n dx$, where n is a positive integer.
- (b) Using Rodrigue's formula, find $P_0(x), P_1(x), P_2(x), P_3(x)$ and hence express $6P_0(x) - 7P_1(x) + 8P_2(x) + 3P_3(x)$ as a polynomial in x.

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6 (a) Find the Laplace transform of the following functions.

(i) $\frac{\sinh t}{t}$ (ii) $te^{-t} \cos t$.

(b) Apply Laplace transforms to solve $y'' + 3y' + 2y = e^{-t}$, $y(0) = 0$, $y'(0) = -1$.7 (a) Find the characteristic equation of the matrix $A = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$ and hence find A^4 .(b) Find the series solution of $(1-x^2)y'' - 2xy' + 6y = 0$ about $x = 0$.

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FACULTY OF ENGINEERING

B.E. II-Semester (AICTE) (Backlog) (New) Examination, February / March 2023

Subject: Mathematics-II

Time: 3 Hours

Max. Marks: 70

- Note: (i) First question is compulsory and answer any four questions from the remaining six questions. Each questions carries 14 Marks.
 (ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.
 (iii) Missing data, if any, may be suitably assumed.

- Determine whether the vectors $(1, -1, 2), (2, 3, 0), (4, 1, 4)$ are linearly dependent.
 - Find the singular solution of the Clairaut's equation $y = x \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2$.
 - Solve $x^2 y'' - 2xy' + 2y = 0$.
 - Prove that $\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$.
 - Find $L\left\{\frac{\sin ht}{t}\right\}$.
 - Find the rank of matrix $A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & -1 & 2 & 3 \\ 2 & 3 & 4 & 5 \end{bmatrix}$.
 - Find the orthogonal trajectories of the family of curves $x^2 - y^2 = c$, where c is a parameter.
- Reduce the quadratic form $Q = 3x^2 + 5y^2 + 3z^2 - 2xy - 2yz + 2zx$ to canonical form and find its nature.
- Solve $(3xy - 2y^2)dx + (x^2 - 2xy)dy = 0$.
 - Solve $(x + 2y^3) \frac{dy}{dx} = y$.
- Find the general solution of $\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 6y = e^{2x}(1+x)$.
 - Using the method of variation of parameters, solve $\frac{d^2 y}{dx^2} + y = x$.
- Prove that $\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$.
 - Find the series solution of $y' - 4y = 0$ about $x = 0$.
- Find $L\{t \sin 2t \cos 4t\}$.
 - Using Laplace transform, solve $\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} - 3y = 0, y(0) = 0, y'(0) = 4$.
- Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ and hence find A^{-1} .
 - Express $3x^3 + x^2 - 2x + 4$ in terms of Legendre polynomials.

FACULTY OF ENGINEERING

**B.E. (Common to all Branches) II Semester (AICTE) (Main & Backlog) Examination,
September / October 2023**

Subject: Mathematics-II**Time: 3 Hours****Max. Marks: 70**

Note: (i) First question is compulsory and answer any four questions from the remaining six questions. Each questions carries 14 Marks.

(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.

(iii) Missing data, if any, may be suitably assumed.

1. (a) If the sum of the eigen values of $A = \begin{bmatrix} 5 & 7 & 3 \\ -2 & k & 5 \\ 0 & 3 & 2 \end{bmatrix}$ is -10 , then find k .

(b) Define an exact differential equation.

(c) Solve $x^2 y'' - 2xy' - 4y = 0$.

(d) Evaluate $\int_0^{\pi/2} \sin^7 \theta \cos^5 \theta d\theta$ using Gamma and Beta functions.

(e) Find $L\{e^{-4t} t^2\}$.

(f) Find the matrix of the quadratic form

$$Q = 2x_1^2 + 4x_2^2 + 5x_3^2 - 6x_1x_2 + 8x_2x_3 - 10x_3x_1.$$

(g) Obtain the singular solution of $y = xy' - \frac{1}{y'}$.

2. (a) Find the rank of the matrix $A = \begin{bmatrix} 2 & -1 & 3 & 2 \\ -4 & 0 & 3 & 5 \\ 7 & 2 & 1 & 1 \end{bmatrix}$ by reducing to echelon form.

(b) Verify Cayley-Hamilton theorem for $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and hence find A^{-1} .

3. (a) Solve $(x^2 + y^3)dx - xy^2dy = 0$.

(b) Find the orthogonal trajectories of the family of circles passing through $(0,2)$ and $(0,-2)$.

4. (a) Solve $\frac{d^2y}{dx^2} + 5 \frac{dy}{dx} + 4y = 3e^{-x} + 2x + \sin x$.

(b) Solve $\frac{d^2y}{dx^2} + y = \tan x$ by the method of variation of parameters.

5. (a) Prove that $\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$.

(b) State Rodrigue's formula and hence find $P_0(x), P_1(x), P_2(x)$ and $P_3(x)$.

6. (a) Find (i) $L\left\{\frac{\sinh t}{t}\right\}$ and (ii) $L\{e^{-t} \sin^2 t\}$

(b) Using Laplace transforms, solve $\frac{d^2y}{dt^2} + 25y = 10 \cos 5t$, $y(0) = 2$, $y'(0) = 0$.

7. (a) Find the values of λ and μ for which the system equations $x + y + z = 3$,
 $x + 2y + 2z = 6$, $x + \lambda y + 3z = \mu$ has (i) no solution (ii) a unique solution and
(iii) infinite number of solutions.

(b) Using convolution theorem, find $L^{-1}\left\{\frac{1}{s(s^2+4)}\right\}$.

FACULTY OF ENGINEERING

B.E. II-Semester (AICTE) (Backlog) (New) Examination, February / March 2023

Subject: Mathematics-II

Time: 3 Hours

Max. Marks: 70

Note: (i) First question is compulsory and answer any four questions from the remaining six questions. Each questions carries 14 Marks.

(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.

(iii) Missing data, if any, may be suitably assumed.

- (a) Determine whether the vectors $(1, -1, 2), (2, 3, 0), (4, 1, 4)$ are linearly dependent.

(b) Find the singular solution of the Clairaut's equation $y = x \frac{dy}{dx} + \left(\frac{dy}{dx}\right)^2$.

(c) Solve $x^2 y'' - 2xy' + 2y = 0$.

(d) Prove that $\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$.

(e) Find $L\left\{\frac{\sin ht}{t}\right\}$.

(f) Find the rank of matrix $A = \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & -1 & 2 & 3 \\ 2 & 3 & 4 & 5 \end{bmatrix}$.

(g) Find the orthogonal trajectories of the family of curves $x^2 - y^2 = c$, where c is a parameter.
- Reduce the quadratic form $Q = 3x^2 + 5y^2 + 3z^2 - 2xy - 2yz + 2zx$ to canonical form and find its nature.
- (a) Solve $(3xy - 2y^2)dx + (x^2 - 2xy)dy = 0$.

(b) Solve $(x + 2y^3) \frac{dy}{dx} = y$.
- (a) Find the general solution of $\frac{d^2 y}{dx^2} - 7 \frac{dy}{dx} + 6y = e^{2x}(1+x)$.

(b) Using the method of variation of parameters, solve $\frac{d^2 y}{dx^2} + y = x$.
- (a) Prove that $\beta(m, n) = \frac{\Gamma m \Gamma n}{\Gamma(m+n)}$.

(b) Find the series solution of $y' - 4y = 0$ about $x = 0$.
- (a) Find $L\{t \sin 2t \cos 4t\}$.

(b) Using Laplace transform, solve $\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} - 3y = 0, y(0) = 0, y'(0) = 4$.
- (a) Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$ and hence find A^{-1} .

(b) Express $3x^3 + x^2 - 2x + 4$ in terms of Legendre polynomials.

FACULTY OF ENGINEERING

B.E. II - Semester (AICTE) (Backlog) (New) Examination, February/ March 2024

Subject: Mathematics-II

Time: 3 Hours

Max. Marks: 70

Note: (i) First question is compulsory and answer any four questions from the remaining six questions. Each question carries 14 Marks.

(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.

(iii) Missing data, if any, may be suitably assumed.

1. a) Find the symmetric matrix A for the quadratic form $x_1^2 + 2ix_1x_2 - 8x_1x_3 + 4ix_2x_3 + 4x_3^2$.
 b) Solve $\cos^2 x \frac{dy}{dx} + y = \tan x$
 c) Solve $(D^3 + 2D^2 + D)y = x^2 e^{2x} + \sin^2 x$.
 d) Show that $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$ and find the value of $\Gamma(7/2)$.
 e) Express $6P_3(x) - 2P_1(x) + P_0(x)$ in terms of powers of x.
 f) Write Cayley -Hamilton theorem and verify it for the matrix $A = \begin{bmatrix} 1 & 4 \\ 3 & 2 \end{bmatrix}$.
 g) Find $L^{-1}\left\{\frac{s+3}{(s-1)(s+2)}\right\}$
2. a) Find the rank of the matrix $A = \begin{bmatrix} 1 & 3 & -4 \\ -1 & -3 & 4 \\ 2 & 6 & -5 \end{bmatrix}$ using elementary row operations.
 b) Find Eigen values and Eigen vectors for the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$
3. a) Find the integrating factor and solve the differential equation
 $y(1 + xy^2)dx + 2(x^2y^2 + x + y^4)dy = 0, y(0) = 1$.
 b) Solve $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 4y = (1 + x)^2$.
4. a) Solve $y'' - 2y' + y = e^x \log x$ by method of variation of parameters.
 b) Find the orthogonal trajectory of family of curves $r = c(1 + \cos \theta)$.
5. a) Solve Legendre's differential equation $(1 - x^2)y'' - 2xy' + n(n + 1)y = 0, n \in Z^+$ and write its General solution.
 b) Write an expression for Legendre's polynomial $P_n(x)$, and find $P_0(x)$, $P_1(x)$ and $P_2(x)$.

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6. a) Test for consistency and solve the following system of equations:

$$2x - 3y + 7z = 5, 3x + y - 3z = 13, 2x + 19y - 47z = 32.$$

b) Reduce the quadratic form $6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$ to canonical form through orthogonal transformation. Find index and signature

7. a) Write Convolution theorem, use it to solve the differential equation

$$y'' + 3y' + 2y = e^{-t}, y(0) = 0, y'(0) = -1$$

b) Apply Laplace transform to solve the initial value problem

$$y''' - 3y'' + 3y' - y = t^2 e^t$$
$$y(0) = 1, y'(0) = 0, y''(0) = -2$$

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FACULTY OF ENGINEERING

B.E. II - Semester (AICTE) (Backlog) (New) Examination, February/ March 2025

Subject: Mathematics - II

Time: 3 Hours

Max. Marks: 70

Note: (i) First question is compulsory and answer any four questions from the remaining six questions. Each question carries 14 Marks.

(ii) Answer to each question must be written at one place only and in the same order as they occur in the question paper.

(iii) Missing data, if any, may be suitably assumed.

1. a) Are the following vectors linearly independent or not $(2, 2, 0), (3, 0, 2), (2, -2, 2)$.
 - b) Solve $(x^2 - ay)dx = (ax - y^2)dy$
 - c) Find the Integrating Factor of the linear differential equation $\frac{dy}{dx} + \frac{x}{1-x^2}y = \frac{ax}{1-x^2}$.
 - d) Check whether the functions $\log x, \log x^n$ are linearly independent or not.
 - e) Evaluate $\int_0^{\infty} \sqrt{x} e^{-x^2} dx$ in terms of Gamma functions.
 - f) Classify the singular points of $(1-x^2)y'' - 2xy' + 2y = 0$
 - g) Find the inverse Laplace transform of $\frac{1}{(s+1)(s+2)}$.
2. a) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -1 \\ 2 & -4 & -4 \end{bmatrix}$. Hence find A^{-1} .
 - b) Find the values of a and b for which the equations $x + y + z = 3, x + 2y + 2z = 6, x + ay + 3z = b$ have
(i) no solution (ii) a unique solution (iii) infinite no. of solutions.
3. a) Show that the one parameter family of curves $y^2 = 4c(c+x)$ are self orthogonal.
 - b) Solve $y(2x^2 - xy + 1)dx + (x - y)dy = 0$.
4. a) Solve $(D^2 - 3D + 2)y = xe^{3x} + \sin 2x$.
 - b) Solve $x^2 y'' - xy' + y = \log x$.
5. a) Prove that $(2n+1)P_n(x) = P'_{n+1}(x) - P'_{n-1}(x)$.
 - b) Show that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

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6. a) Find $L\left[\int_0^t t e^{-t} \sin 4t dt\right]$

b) Solve the differential equation using the Laplace transform. $(D^2 + 5D + 6)x = 5e^t$. Given that $x(0) = 2$ and $x'(0) = 1$.

7. a) Reduce the quadratic form $3x^2 - 2y^2 - z^2 - 4xy + 8xz + 12yz$ to canonical form by orthogonal transformation. Write also nature and index.

b) It is known that $\frac{1}{x}$ is a solution of the differential equation $x^2 y'' + 4xy' + 2y = 0$. Find the second linearly independent solution and write the general solution.

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